



EEE 1101

Electrical Circuits

Md. Mohin Uddin

Lecturer, Dept. of EEE

DUET, Gazipur-1707

Email: mohineee@duet.ac.bd

Phone: 01571-501559

Room no.: 113



Course Content

Basic Circuit Concept:

Basic electrical elements and quantities, Series and parallel circuits, Circuit Laws and Theorems, Mesh and Node analysis, Circuit with dependent sources, Capacitors and Inductors, Fundamental of AC Circuit, Phasor Algebra, AC circuit analysis, Series and parallel resonance, Influence of Circuit Parameters on Resonant Behavior, Q of a circuit, Wave trap, Series-Parallel Tuning.

Balanced Polyphase Circuits:

Two and four phase system, Three-phase four-wire system, Three-phase three-wire system, N-phase star and mesh, Balanced wye and delta loads, Three origin vector diagram of a balanced three phase system, Power measurement (both active and reactive) in balanced systems, Copper required to transmit power under fixed conditions.

Unbalanced Polyphase Circuits:

Unbalanced wye and delta loads, Network solutions, Phase-sequence effects, Methods of checking voltage phase sequence, Power measurements (both active and reactive) in unbalanced three phase systems.



Reference Books

1. Fundamentals of Electric Circuits (5th edition)

- Charles K. Alexander
- Matthew N. O. Sadiku

2. Electric Circuits (10th edition)

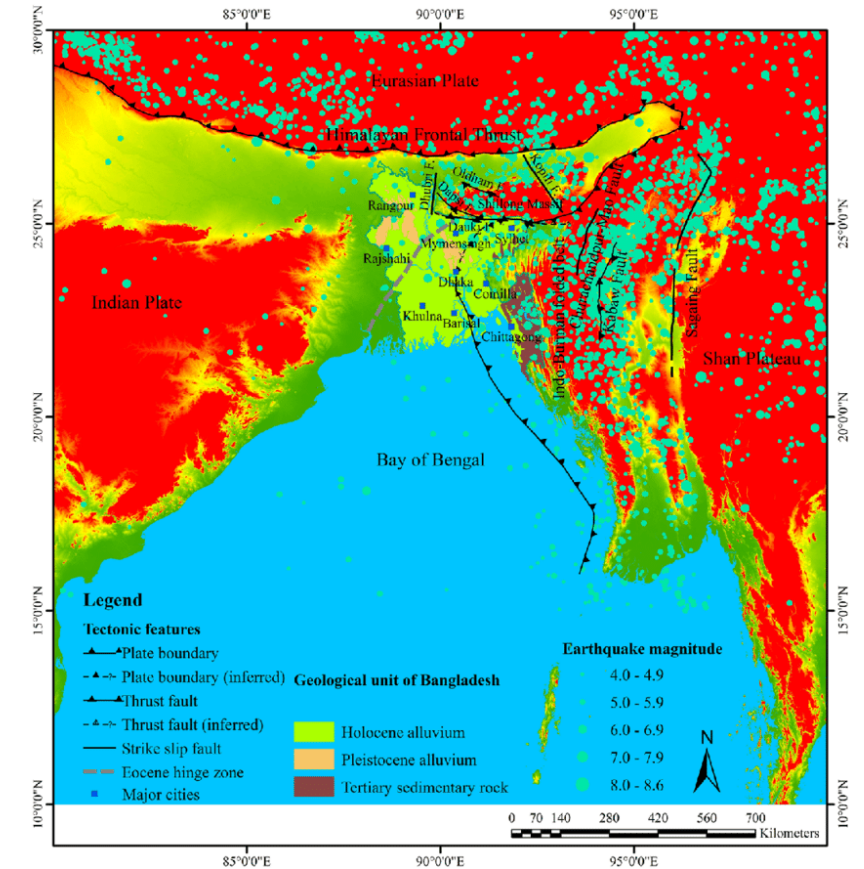
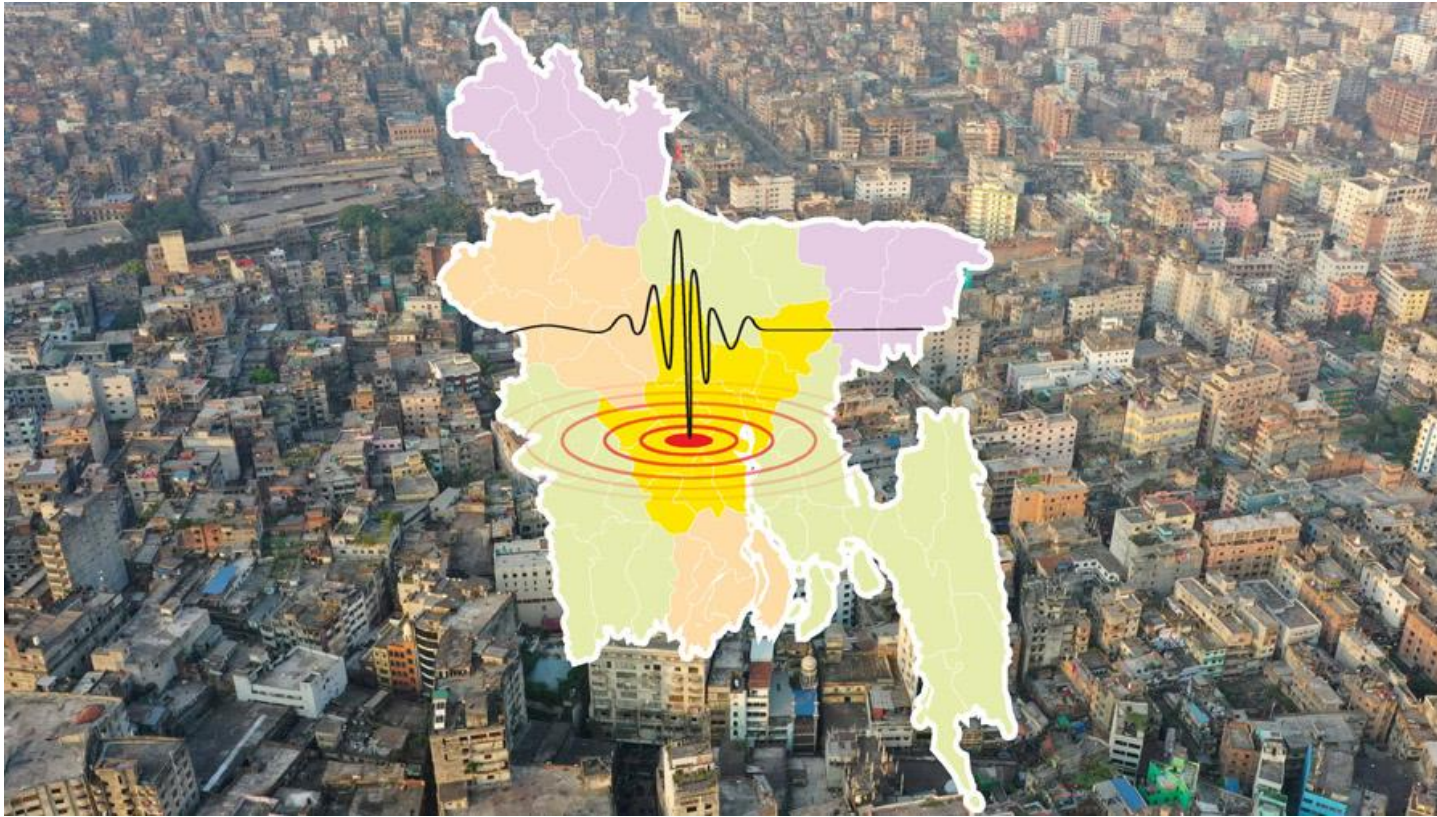
- James W. Nilsson
- Susan A. Riedel

3. Alternating Current Circuits (4th edition)

- Russel M. Kerchner
- George F. Corcoran

4. Introductory Circuit Analysis (11th edition)

- Robert L. Boylestad



Natural disasters remind us that human power is limited but Allah's control is absolute.

**"Whoever relies upon Allah – He is sufficient for him."
— (Surah At-Talaq: 3)**

Never lose hope. Allah is aware of every situation we face and is making the best decisions for us. So, even in moments of sorrow, place your complete trust in Him.

Mechanical Resonance



Tacoma narrows bridge collapse

Four months after the opening of the first Tacoma Narrows Bridge, on the morning of November 7, 1940, it suffered collapse in a wind of about 42 miles (67 km) per hour.

Series Resonance



Series Resonance: A series circuit containing R, L, and C is in resonance when the resultant reactance is zero. Since the drop across the inductance leads the current by 90° whereas that across the condenser lags the current by 90° , the two drops are opposite. If they are made equal as in Fig.2, the reactive voltage drops neutralize and the impressed voltage is equal only to the resistance drop. This condition is called series resonance. Inspection of the vector diagram of Fig. 2 shows that the applied voltage is in phase with the current. The power factor is unity, and the circuit is in resonance. Thus for series resonance,

$$IX_L = IX_C \quad \text{or} \quad X_L = X_C$$

The series resonant frequency is

$$f_s = \frac{1}{2\pi\sqrt{LC}}$$

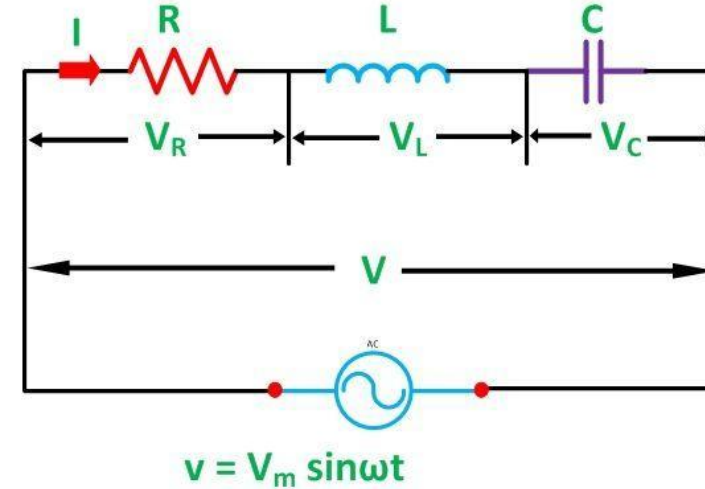


Fig.1

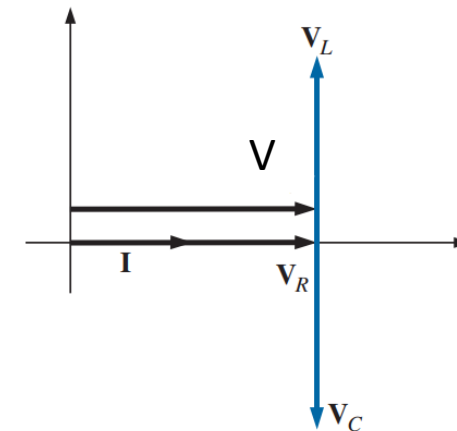


Fig.2

Phasor diagram for the series resonant circuit at resonance.

Series Resonance



The current is

$$I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + (X_L - X_C)^2}} = \frac{V}{\sqrt{R^2 + \left(2\pi fL - \frac{1}{2\pi fC}\right)^2}}$$

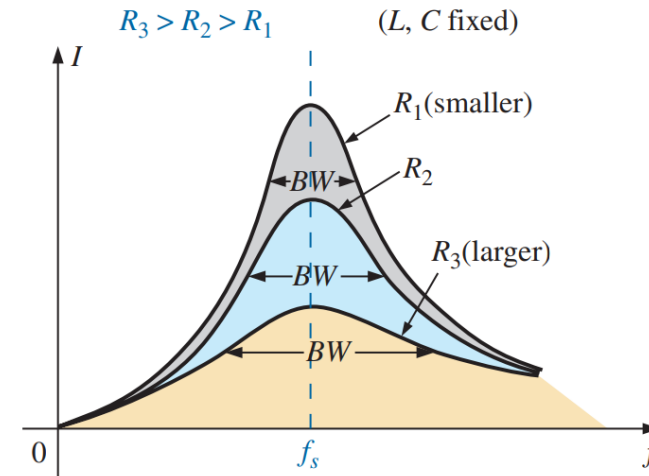
The voltage drop across the resistance, inductance and capacitance is

$$V_R = IR = \frac{VR}{\sqrt{R^2 + \left(2\pi fL - \frac{1}{2\pi fC}\right)^2}}$$

$$V_L = IX_L = \frac{VX_L}{\sqrt{R^2 + \left(2\pi fL - \frac{1}{2\pi fC}\right)^2}}$$

$$V_C = IX_C = \frac{VX_C}{\sqrt{R^2 + \left(2\pi fL - \frac{1}{2\pi fC}\right)^2}}$$

Effect of resistance on current variation in the range of series resonance:



In a series resonant circuit, the resistor controls how sharp or wide the response is. If you increase the resistance, the peak current gets smaller and the bandwidth becomes wider, making the curve broader. On the other hand, if you decrease the resistance, the peak current gets larger and the bandwidth becomes narrower, resulting in a sharper and more selective curve.

Series Resonance



The Effect of Varying Inductance (L) on Series Resonance

When the inductance (L) in a series resonant circuit is varied, the voltage across the capacitor (V_C) and the inductor (V_L) behave differently. The maximum value of V_C occurs exactly at the resonant frequency because it depends directly on the current (I), which is also at its maximum at resonance. In contrast, the maximum voltage across the inductor (V_L) occurs at a frequency after (higher than) resonance. This is because V_L is the product of both the current (I) and the inductive reactance (X_L), which increases with frequency. Even after the peak current passes, the increasing X_L continues to push V_L higher until the point where the rate of decrease in current finally outweighs the rate of increase in reactance.

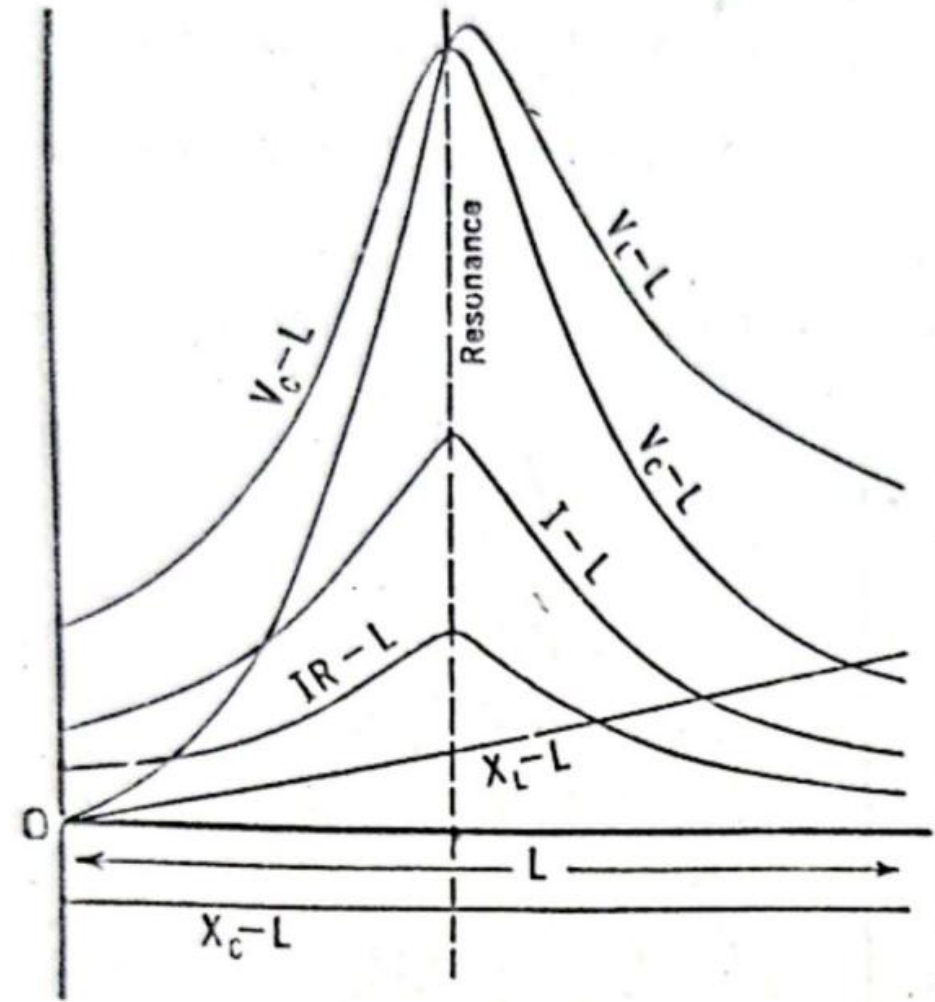


FIG. Series resonance by varying L .

Series Resonance



This specific point for maximum V_L can be found mathematically by setting the derivative dV_L/dX_L to zero.

$$\frac{dV_L}{dX_L} = \frac{[R^2 + (X_L - X_C)^2]^{-1/2} V - V X_L \frac{1}{2} [R^2 + (X_L - X_C)^2]^{-3/2} 2(X_L - X_C)}{R^2 + (X_L - X_C)^2} = 0$$

and
$$X_L = \frac{R^2 + X_C^2}{X_C}$$

or
$$L = \frac{1}{2\pi f} \left(\frac{R^2 + X_C^2}{X_C} \right) = C(R^2 + X_C^2)$$

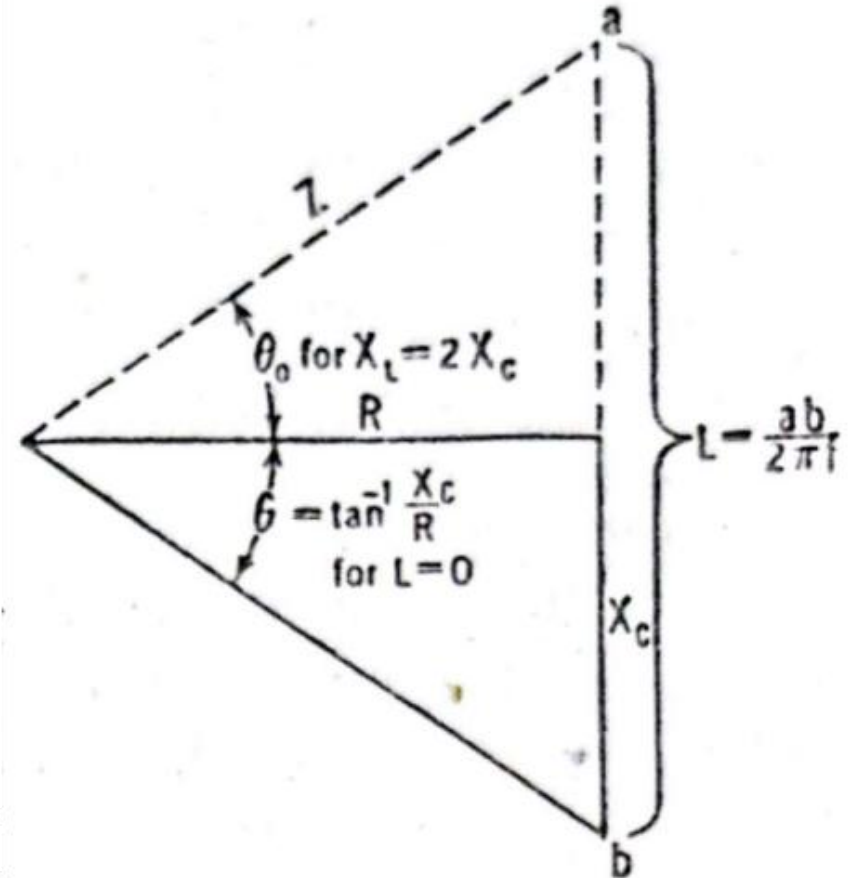


FIG. Impedance diagram showing the power factor angle θ as L is varied in an RLC series circuit.

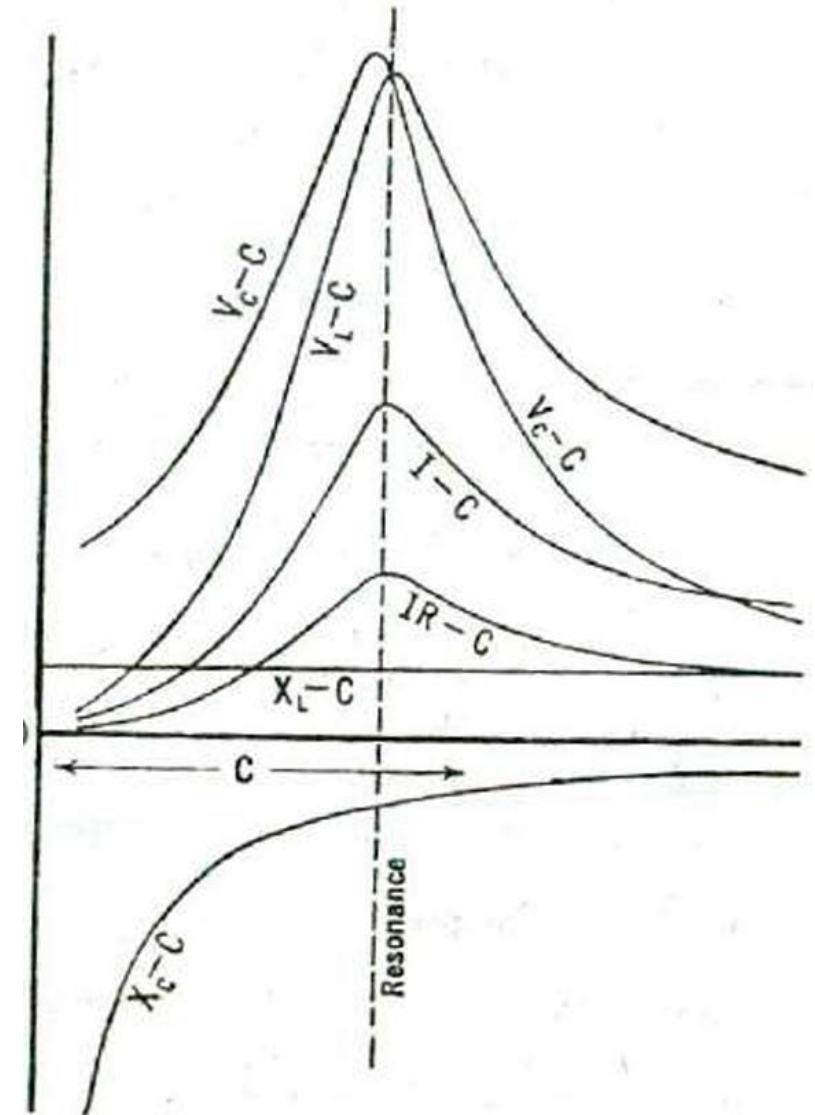
Series Resonance



The Effect of Varying Capacitance (C) on Series Resonance:

Here the drop across the inductance is a maximum when the current is a maximum, since X_L is constant. The maximum drop across the condenser occurs before resonance. At resonance, X_C is decreasing whereas the current is not changing (slope being zero). The drop IX_C must, therefore, be decreasing. Consequently, the drop must have been a maximum before resonance.

At resonance the drops across the inductance and the capacitance are equal and opposite.



Series Resonance



This specific point for maximum V_C can be found mathematically by setting the derivative dV_C/dX_C to zero.

$$\begin{aligned} \frac{dV_C}{dX_C} &= 0 \\ \Rightarrow \frac{d}{dX_C} \left[\frac{VX_C}{\sqrt{R^2 + (X_L - X_C)^2}} \right] &= 0 \\ \Rightarrow \frac{d}{dX_C} V \left[\frac{\sqrt{R^2 + (X_L - X_C)^2} \cdot 1 - X_C \cdot \frac{1}{2} \cdot \frac{2(X_L - X_C)(-1)}{\sqrt{R^2 + (X_L - X_C)^2}}}{(\sqrt{R^2 + (X_L - X_C)^2})^2} \right] &= 0 \\ \Rightarrow \sqrt{R^2 + (X_L - X_C)^2} + \frac{X_L X_C - X_C^2}{\sqrt{R^2 + (X_L - X_C)^2}} &= 0 \end{aligned}$$

$$\Rightarrow R^2 + (X_L - X_C)^2 = X_C^2 - X_L X_C$$

$$\Rightarrow R^2 + X_L^2 + X_C^2 - 2X_L X_C = X_C^2 - X_L X_C$$

$$\Rightarrow R^2 + X_L^2 = X_L X_C$$

$$\Rightarrow X_C = \frac{R^2 + X_L^2}{X_L}$$

$$\Rightarrow \frac{1}{\omega C} = \frac{R^2 + X_L^2}{\omega L}$$

$$\Rightarrow \frac{L}{R^2 + X_L^2} = C$$

$$\Rightarrow C = \frac{L}{R^2 + X_L^2}$$

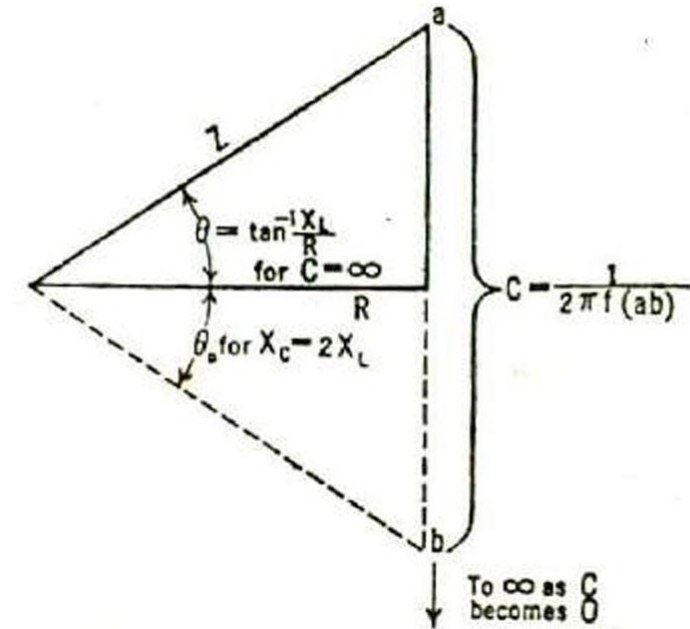


Figure: Impedance diagram indicating range of power factor angle θ as C varied in an RLC series circuit

Homework:

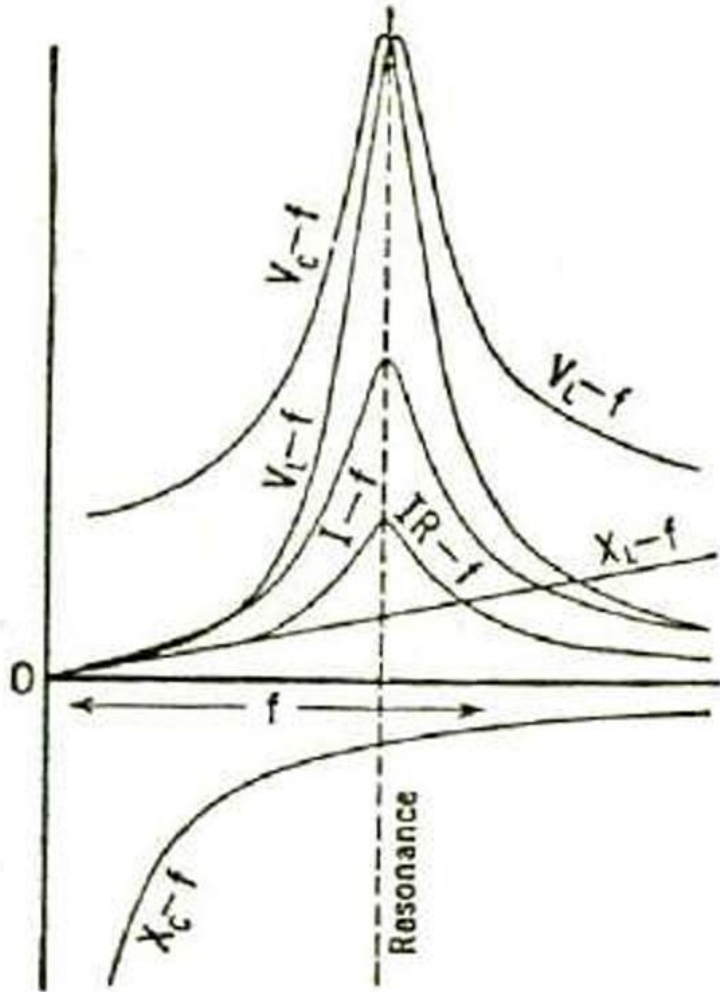
A series RLC circuit operates at frequency f .

- With capacitance C fixed and inductance L varying, find the value of inductance L that produces the maximum voltage across the inductor.
- With inductance L fixed and capacitance C varying, find the value of capacitance C that produces the maximum voltage across the capacitor.

Series Resonance



The Effect of Varying frequency (f) on Series Resonance:



- At the resonant frequency, neither the inductor nor the capacitor has the maximum voltage drop across it.
- This method of achieving resonance is a combination of two other methods previously discussed.
- The current in the circuit is zero at both extremely low (zero) and extremely high (infinite) frequencies.

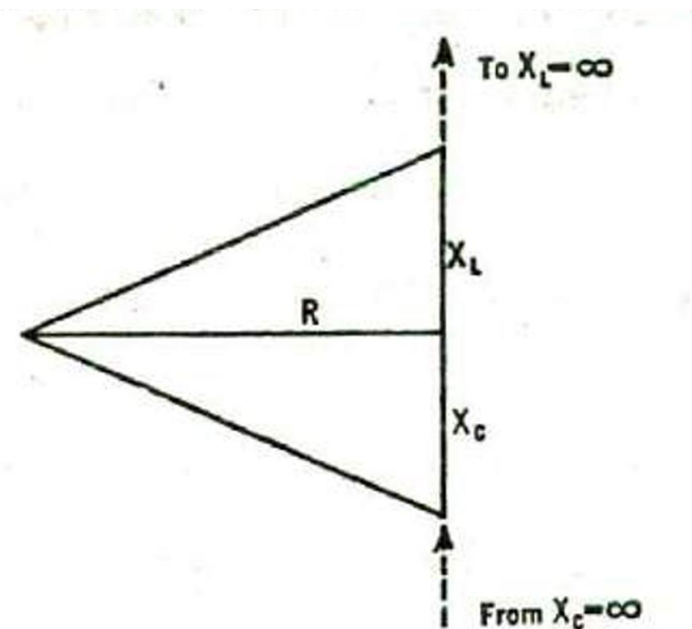


Figure: Impedance triangle indicating variation of phase angle from -90 to $+90$ as frequency is varied in an RLC series circuit

The Series RLC Circuit as a Selector:

The RLC circuit passes frequencies near the resonant frequency more readily than other frequencies. The circuit thus has selective properties. The band of frequencies which is passed quite readily is called the pass band. The pass band is sometimes arbitrarily considered to be the range of frequency over which the current is equal to or greater than $V/(\sqrt{2} R)$, as indicated in Figure.

The current flow in a RLC series circuit,

$$I = \frac{V}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

For the half-power points the current flow,

$$I = \frac{V}{\sqrt{2}R}$$

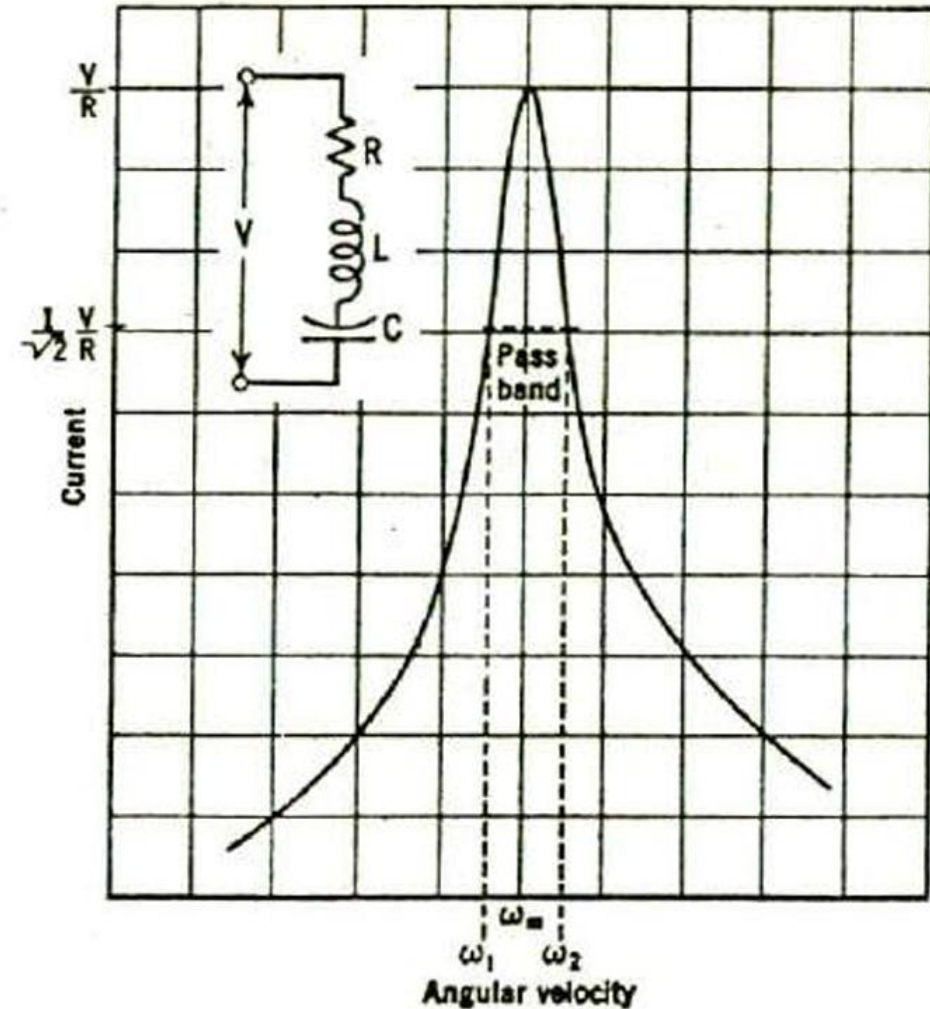


Figure: The RLC series branch, as a band selector.

Series Resonance



Therefore,

$$\frac{V}{\sqrt{2}R} = \frac{V}{\sqrt{R^2 + (\omega_n L - \frac{1}{\omega_n C})^2}}$$

$$\Rightarrow 2R^2 = R^2 + (\omega_n L - \frac{1}{\omega_n C})^2$$

$$\Rightarrow R = \pm (\omega_n L - \frac{1}{\omega_n C})$$

$$\Rightarrow \omega_n R = \pm (\omega_n L - \frac{1}{C})$$

Taking positive sign only,

$$\omega_n R = \omega_n L - \frac{1}{C}$$

$$\Rightarrow \omega_n L - \omega_n R - \frac{1}{C} = 0$$

$$\Rightarrow \omega_n = \frac{R \pm \sqrt{R^2 + 4L/C}}{2L}$$

$$\Rightarrow \omega_n = \frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

If we take negative sign,

$$\omega_n = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

In general,

$$\omega_n = \pm \frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

In a selective RLC series circuit,

$$\left(\frac{R}{2L}\right)^2 \ll \frac{1}{LC}$$

neglecting $\left(\frac{R}{2L}\right)^2$, we obtain

$$\omega_n = \pm \frac{R}{2L} \pm \sqrt{\frac{1}{LC}}$$

$$\Rightarrow \omega_n = \pm \frac{R}{2L} \pm \omega_m$$

Series Resonance



If only positive value of ω_m is considered,

$$\omega_n = \pm \frac{R}{2L} + \omega_m = \omega_m \pm \frac{R}{2L}$$

$$\therefore \boxed{\omega_1 = \omega_m - \frac{R}{2L}} \text{ and } \boxed{\omega_2 = \omega_m + \frac{R}{2L}}$$

The bandwidth of the selector,

$$\Delta\omega = \omega_2 - \omega_1 = \left(\omega_m + \frac{R}{2L}\right) - \left(\omega_m - \frac{R}{2L}\right)$$

$$\Rightarrow \Delta\omega = \frac{R}{L}$$

$$\Rightarrow \boxed{\Delta f = \frac{R}{2\pi L}}$$

Note: Those frequencies corresponding to 0.707 of the maximum current are called **the band frequencies, cut off frequencies, half-power frequencies, or corner frequencies.**

Series Resonance



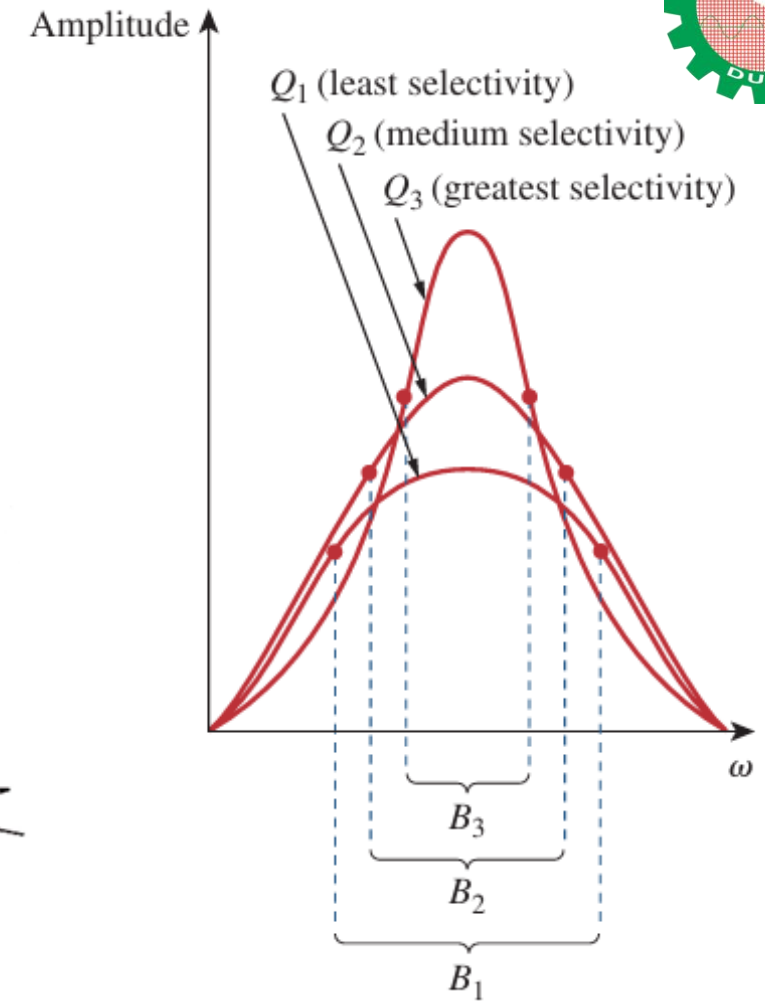
The Q of a series Circuit :

The **quality factor** of a resonant circuit is the ratio of its resonant frequency to its bandwidth.

In the case of the series RLC circuit,

$$Q_s = \frac{\omega_m}{\Delta\omega} = \frac{f_m}{\Delta f} = \frac{\frac{1}{2\pi\sqrt{LC}}}{\frac{R}{2\pi L}} = \frac{L}{R\sqrt{LC}} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$Q_s = \frac{L}{R\sqrt{LC}} = \frac{L}{R} \cdot \frac{1}{\sqrt{LC}} = \frac{L}{R} \cdot \omega_m = \frac{\omega_m L}{R} = \frac{X_L}{R}$$



A small Q_s is associated with a resonant curve having a large bandwidth and a small selectivity, while a large Q_s indicates the opposite.

For circuits where $Q_s \geq 10$, a widely accepted approximation is that the resonant frequency bisects the bandwidth and that the resonant curve is symmetrical about the resonant frequency.

Locus or Circle diagram of Series Circuit

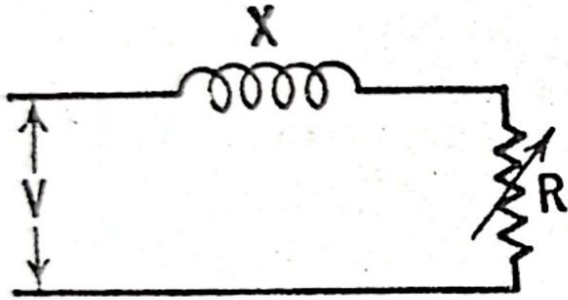


Figure (a): Series circuit with variable R .

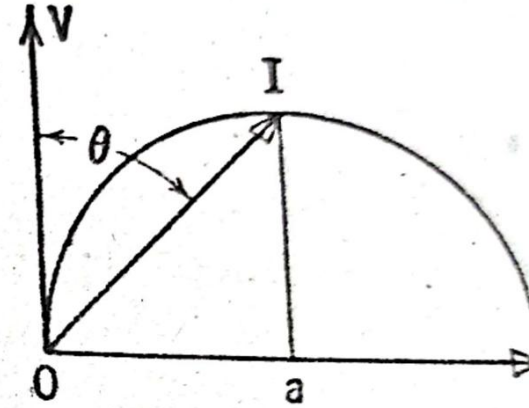
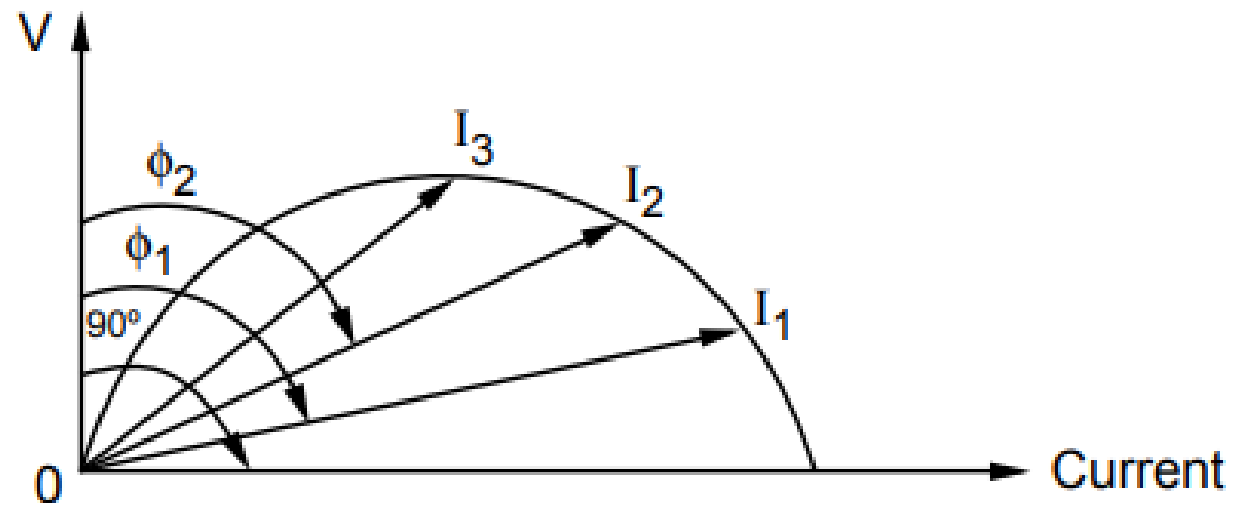
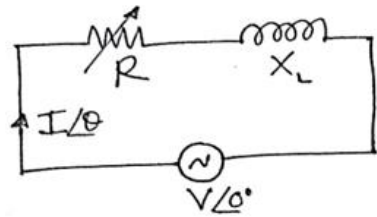


Figure (b): Circle diagram of Figure (a) for constant V and X but with variable R .

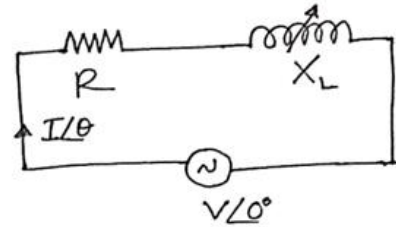
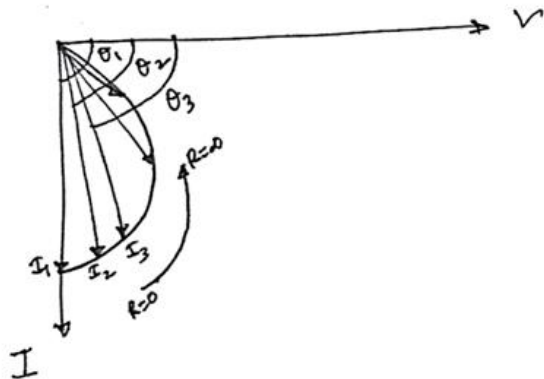


Locus or Circle diagram of Series Circuit



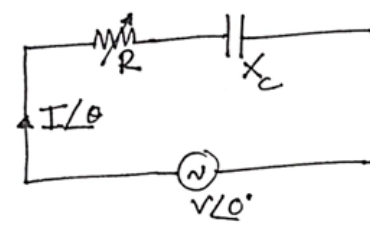
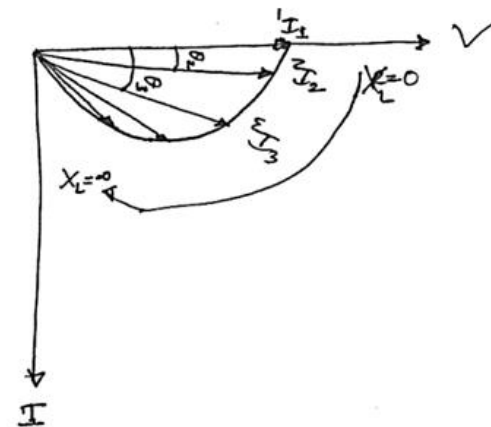
Boundary Condition:

$R=0$	$I = \frac{V}{X_L}$	$\theta = -90^\circ$
$R=\infty$	$I=0$	$\theta = 0^\circ$



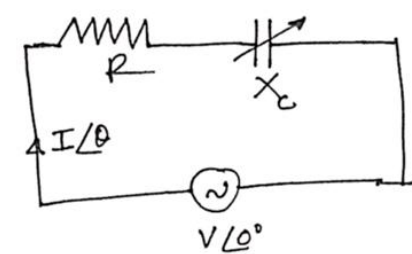
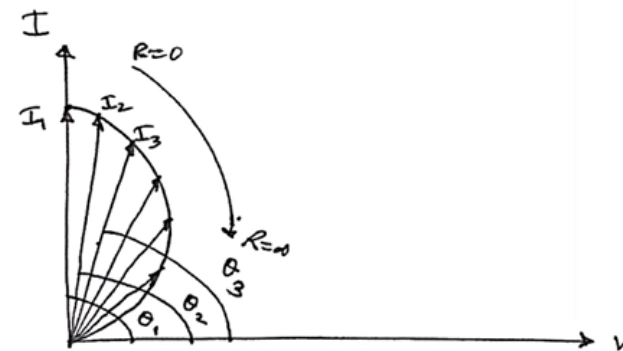
Boundary Condition:

$X_L=0$	$I = \frac{V}{R}$	$\theta = 0^\circ$
$X_L=\infty$	$I=0$	$\theta = -90^\circ$



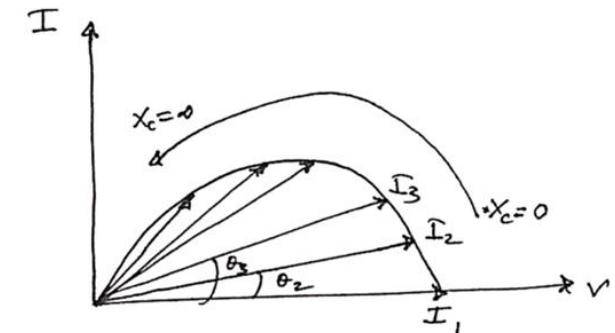
Boundary Condition:

$R=0$	$I = \frac{V}{X_C}$	$\theta = +90^\circ$
$R=\infty$	$I=0$	$\theta = 0^\circ$



Boundary Condition:

$X_C=0$	$I = \frac{V}{R}$	$\theta = 0^\circ$
$X_C=\infty$	$I=0$	$\theta = +90^\circ$



Conversion of Series Impedance to Parallel Branch

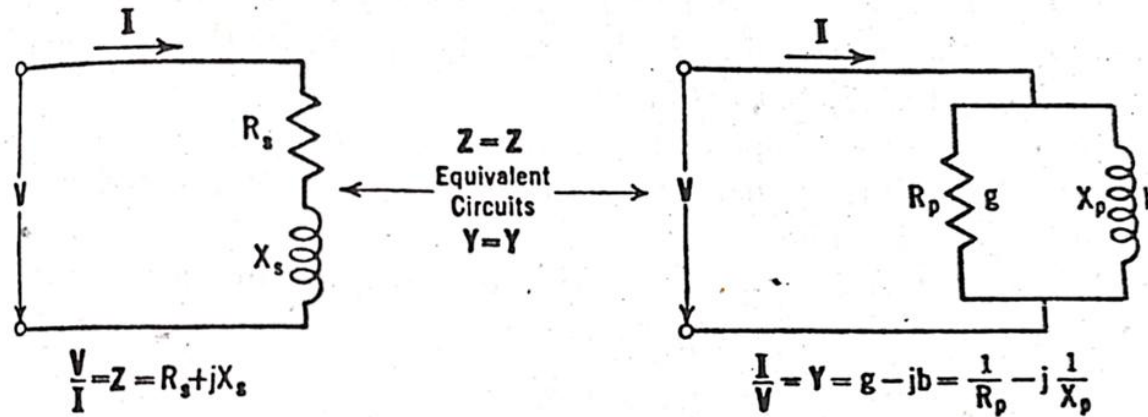
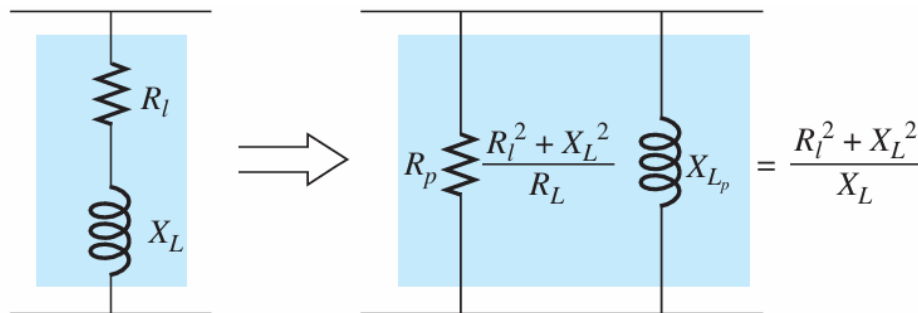


FIG. 22. The parallel equivalent of a series impedance, $R_s + jX_s$.

$$\therefore R_p = \frac{R_s^2 + X_s^2}{R_s} \quad \text{and} \quad X_p = \frac{R_s^2 + X_s^2}{X_s}$$



$$Z_s = Z_p$$

$$\Rightarrow \frac{1}{Z_s} = \frac{1}{Z_p}$$

$$\Rightarrow \frac{1}{R_s + jX_s} = \frac{1}{R_p} + \frac{1}{jX_p}$$

$$\Rightarrow \frac{R_s - jX_s}{(R_s + jX_s)(R_s - jX_s)} = \frac{1}{R_p} + \frac{1}{jX_p}$$

$$\Rightarrow \frac{R_s - jX_s}{R_s^2 - j^2 X_s^2} = \frac{1}{R_p} - j \frac{1}{X_p}$$

$$\Rightarrow \frac{R_s}{R_s^2 + X_s^2} - j \frac{X_s}{R_s^2 + X_s^2} = \frac{1}{R_p} - j \frac{1}{X_p}$$

$$\Rightarrow \frac{1}{\frac{R_s^2 + X_s^2}{R_s}} - j \frac{1}{\frac{R_s^2 + X_s^2}{X_s}} = \frac{1}{R_p} - j \frac{1}{X_p}$$

Parallel Resonance

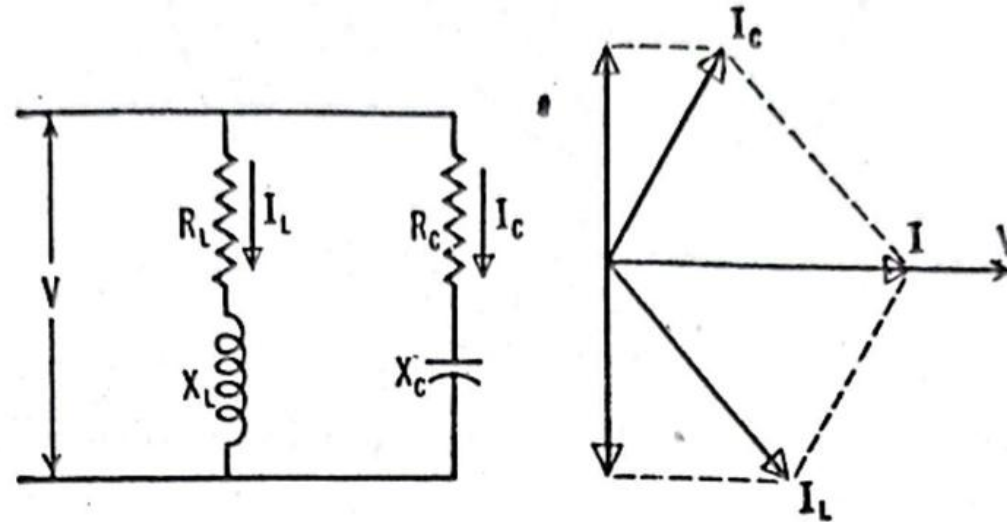


FIG. 24. Circuit and corresponding vector diagram for parallel resonance.

Condition: Resonance occurs in a parallel RLC circuit when the inductive susceptance equals the capacitive susceptance ($b_L = b_C$).

Consequence: This equality means the reactive currents in the inductive and capacitive branches cancel each other out. Therefore, the resultant current from the source is in phase with the applied voltage.

Power Factor: The overall circuit operates at a unity power factor. This specific condition is also called unity-power-factor resonance.

Current at Resonance: It is often inferred that the resultant current is minimum at resonance.

Parallel Resonance



$$\frac{2\pi fL}{R_L^2 + (2\pi fL)^2} = \frac{\frac{1}{2\pi fC}}{R_C^2 + \left(\frac{1}{2\pi fC}\right)^2}$$

The quantities that may be varied are L , C , f , R_L , or R_C .

$$Y = Y_1 + Y_2 = \frac{1}{Z_1} + \frac{1}{Z_2} = \frac{1}{R_L + jX_L} + \frac{1}{R_C - jX_C}$$

$$= \frac{R_L - jX_L}{R_L^2 + X_L^2} + \frac{R_C + jX_C}{R_C^2 + X_C^2}$$

$$= \left(\frac{R_L}{R_L^2 + X_L^2} + \frac{R_C}{R_C^2 + X_C^2} \right) + j \left(\frac{X_C}{R_C^2 + X_C^2} - \frac{X_L}{R_L^2 + X_L^2} \right)$$

At resonance,

$$b_L = b_C$$

$$\Rightarrow \frac{X_L}{R_L^2 + X_L^2} = \frac{X_C}{R_C^2 + X_C^2}$$

$$\Rightarrow \frac{2\pi fL}{R_L^2 + (2\pi fL)^2} = \frac{\frac{1}{2\pi fC}}{R_C^2 + \left(\frac{1}{2\pi fC}\right)^2}$$

Parallel Resonance



The Effect of Varying Inductance (L) on Parallel Resonance:

Current Loci: When L is varied, the inductive branch current (I_L) traces a semicircle on a phasor diagram, with its maximum (diameter) equal to V/R_L .

Resultant Current Locus: The total (resultant) current from the source, found by adding the constant capacitive current (I_C) to the variable I_L , also traces a circle.

Resonance vs. Minimum Current:

- Resonance (unity power factor) occurs at the points where the total current phasor is in phase with the voltage.
- Minimum current occurs at a different point where the resultant current phasor is normal (perpendicular) to its circular locus.
- Therefore, resonance does not necessarily coincide with minimum current in this practical circuit with resistance.

Conditions for Resonance:

- No resonance possible if $V/(2 R_L) < I_C \sin \theta_c$
- One resonant point if $V/(2 R_L) = I_C \sin \theta_c$
- Two resonant points $V/(2 R_L) > I_C \sin \theta_c$

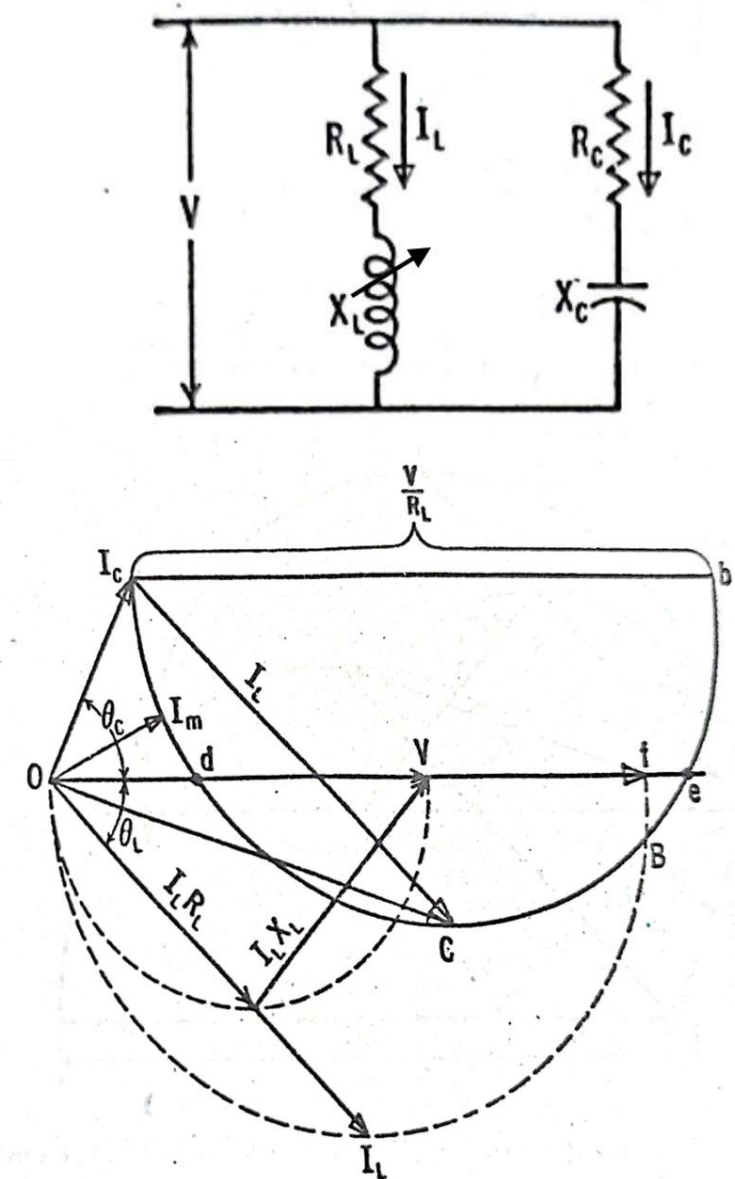


Figure: Locus of the resultant current as L is varied.

Parallel Resonance



The Effect of Varying Capacitance (C) on parallel Resonance:

Current Loci: When C is varied, the capacitive branch current (I_C) traces a semicircle on a phasor diagram, with its maximum (diameter) equal to V/R_C .

Resultant Current Locus: The total (resultant) current from the source, found by adding the constant inductive current (I_L) to the variable I_C , also traces a circle.

Resonance vs. Minimum Current:

- Resonance (unity power factor) occurs at the points where the total current phasor is in phase with the voltage.
- Minimum current occurs at a different point where the resultant current phasor is normal (perpendicular) to its circular locus.
- Therefore, resonance does not necessarily coincide with minimum current in this practical circuit with resistance.

Conditions for Resonance:

- No resonance possible if $V/(2 R_C) < I_L \sin \theta_L$
- One resonant point if $V/(2 R_C) = I_L \sin \theta_L$
- Two resonant points $V/(2 R_C) > I_L \sin \theta_L$

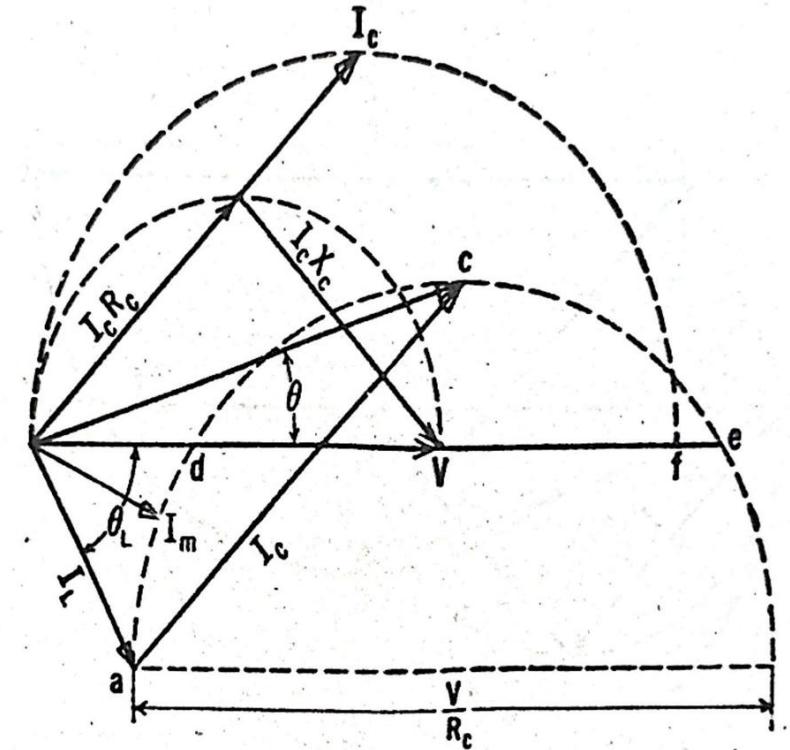
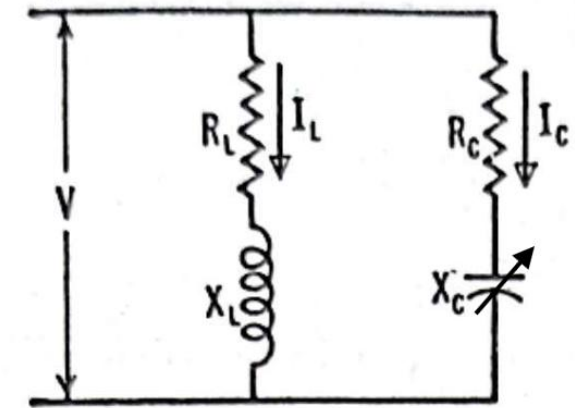
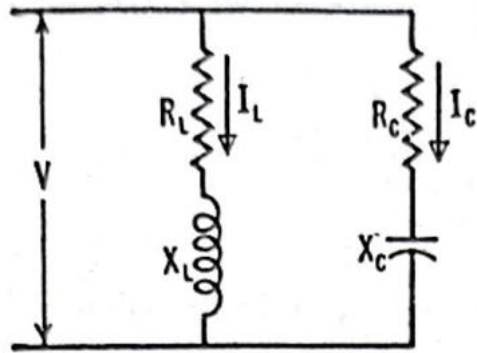


Figure: Locus of the resultant current as C is varied.

Parallel Resonance



The Effect of Varying frequency (f) on parallel Resonance:



The frequency for parallel resonance,

$$f_m = \frac{1}{2\pi\sqrt{LC}} \left[\frac{R_L^2 C - L}{R_C^2 C - L} \right]^{1/2}$$

If $CR_L^2 > L$ and $CR_C^2 < L$,

no real frequency will yield resonance.

If $R_L = R_C$, $f_m = \frac{1}{2\pi\sqrt{LC}}$

If $R_L = R_C = 0$, $f_m = \frac{1}{2\pi\sqrt{LC}}$

$$b_L = b_C$$

$$\frac{X_L}{R_L^2 + X_L^2} = \frac{X_C}{R_C^2 + X_C^2} \Rightarrow \frac{R_L^2 + X_L^2}{X_L} = \frac{R_C^2 + X_C^2}{X_C}$$

$$\Rightarrow \frac{R_L^2}{X_L} + X_L = \frac{R_C^2}{X_C} + X_C \Rightarrow \frac{R_L^2}{2\pi f L} + 2\pi f L = \frac{R_C^2}{(1/2\pi f C)} + \frac{1}{2\pi f C}$$

$$\Rightarrow \frac{R_L^2}{2\pi f L} - \frac{1}{2\pi f C} = 2\pi f C R_C^2 - 2\pi f L$$

$$\Rightarrow 2\pi f [C R_C^2 - L] = \frac{1}{2\pi f} \left[\frac{R_L^2}{L} - \frac{1}{C} \right]$$

$$\Rightarrow 2\pi f [C R_C^2 - L] = \frac{1}{2\pi f L C} [C R_L^2 - L]$$

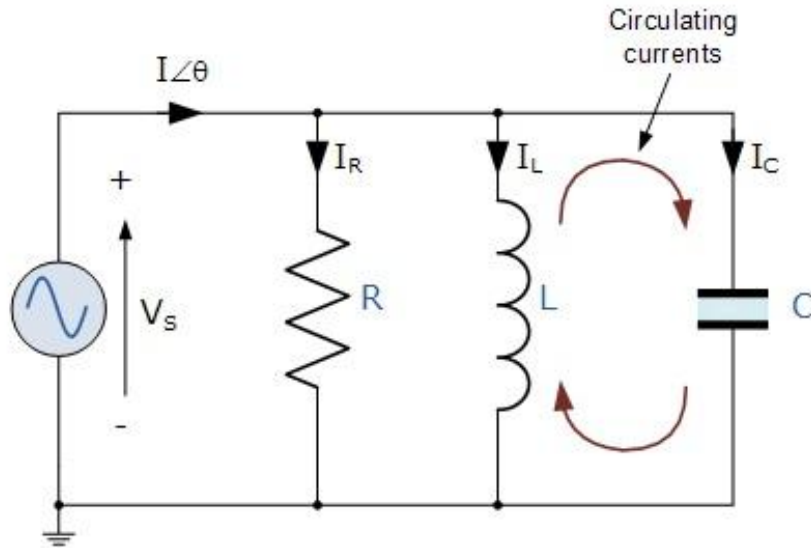
$$\Rightarrow (2\pi f)^2 = \frac{1}{LC} \left[\frac{C R_L^2 - L}{C R_C^2 - L} \right]$$

$$\Rightarrow f = \frac{1}{2\pi\sqrt{LC}} \left[\frac{C R_L^2 - L}{C R_C^2 - L} \right]^{1/2}$$

Parallel Resonance



The Effect of Varying frequency (f) on parallel Resonance:



$$Y = G + j(B_C - B_L)$$

magnitude of Y

$$Y = \sqrt{G^2 + (B_C - B_L)^2}$$

$$= \sqrt{G^2 + \left(2\pi f C - \frac{1}{2\pi f L}\right)^2}$$

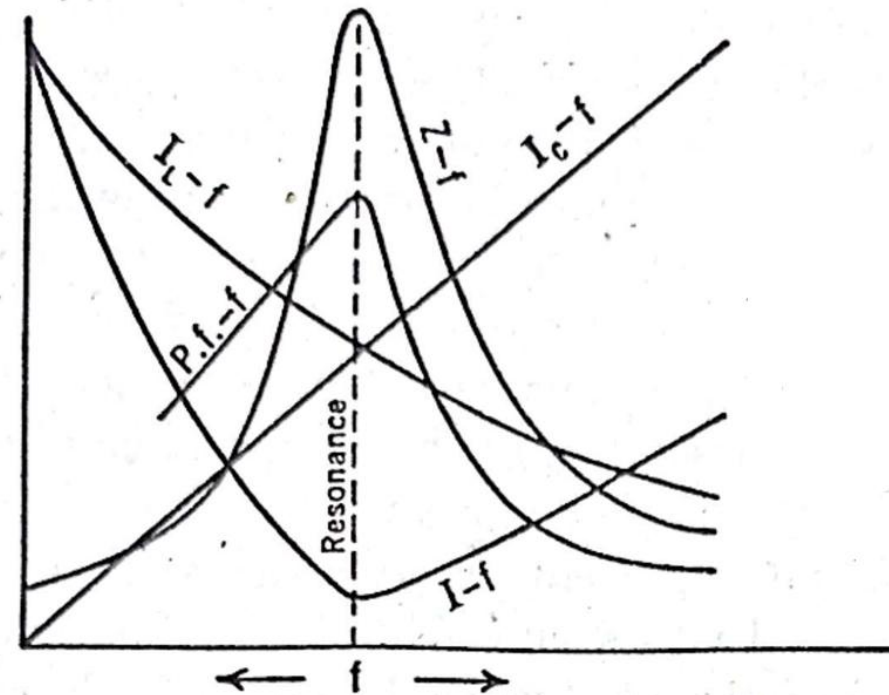


FIG. 28. Parallel resonance by varying frequency.

Parallel Resonance

A simple form Effective band eliminator, suppressor, or wave Trap:

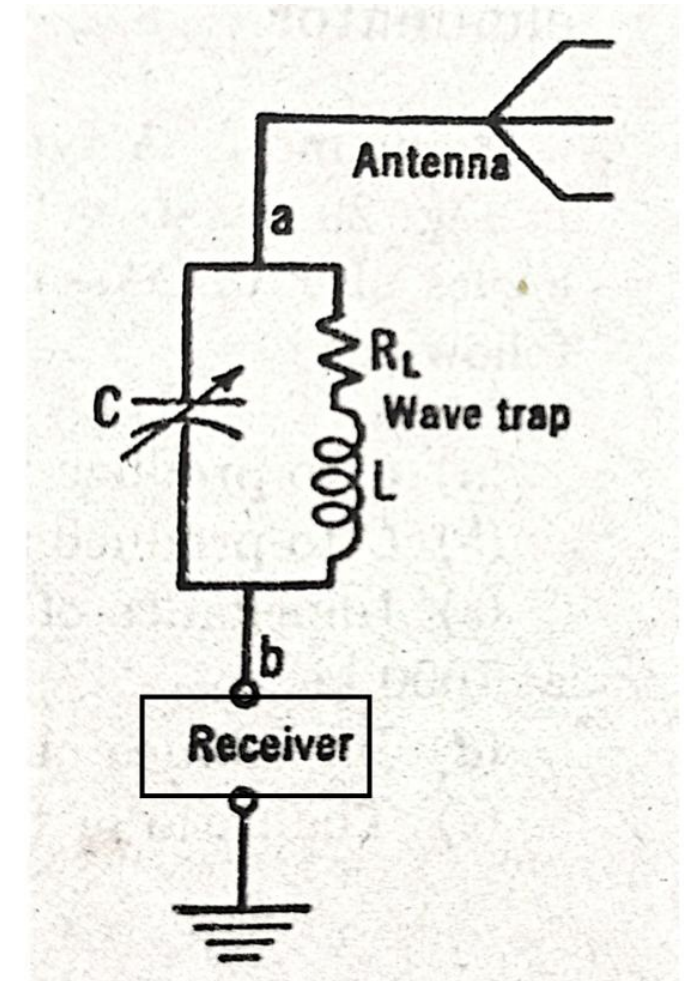
A wave trap is a parallel resonant circuit made of an inductor (L), capacitor (C), and small resistance (R_L). It acts as a band eliminator or suppressor, blocking a specific frequency.

A wave trap operates on the principle of resonance. At its specific resonant frequency, the combined circuit presents a very high impedance. This high impedance effectively blocks signals at that specific frequency from passing through the branch where the trap is placed. Meanwhile, signals at other frequencies see a low impedance path and are allowed to pass with minimal effect.

In system design, a wave trap acts as a notch filter. It is strategically placed within a circuit, often in series with a signal path or across a transmission line, to eliminate a specific unwanted frequency.

Resonant frequency,
$$f_m = \frac{1}{2\pi\sqrt{LC}}$$

Impedance at resonance,
$$Z_m = \frac{L}{CR_L}$$



Parallel Resonance



The admittance,

$$Y = \frac{1}{R_L + jX_L} + \frac{1}{-jX_C} = \frac{R_L}{R_L^2 + X_L^2} - j \frac{X_L}{R_L^2 + X_L^2} + j \frac{1}{X_C}$$

$$= \frac{R_L}{R_L^2 + X_L^2} + j \left(\frac{1}{X_C} - \frac{X_L}{R_L^2 + X_L^2} \right)$$

At Resonance,

$$\frac{1}{X_C} = \frac{X_L}{R_L^2 + X_L^2}$$

$$\Rightarrow \frac{1}{X_C} \cong \frac{X_L}{X_L^2}$$

$$\Rightarrow \frac{1}{X_C} \cong \frac{1}{X_L}$$

$$\Rightarrow 2\pi f C \cong \frac{1}{2\pi f L}$$

$$\Rightarrow f \cong \frac{1}{2\pi \sqrt{LC}}$$

and,

$$Y = \frac{R_L}{R_L^2 + X_L^2}$$

since $R_L^2 \ll (\omega L)^2$,

$$Y \cong \frac{R_L}{X_L^2}$$

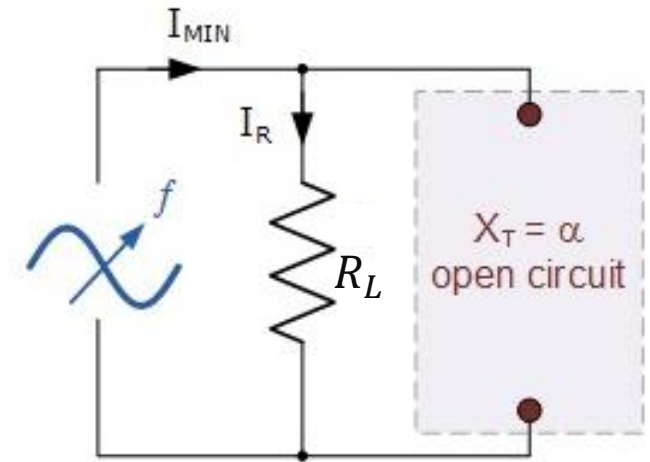
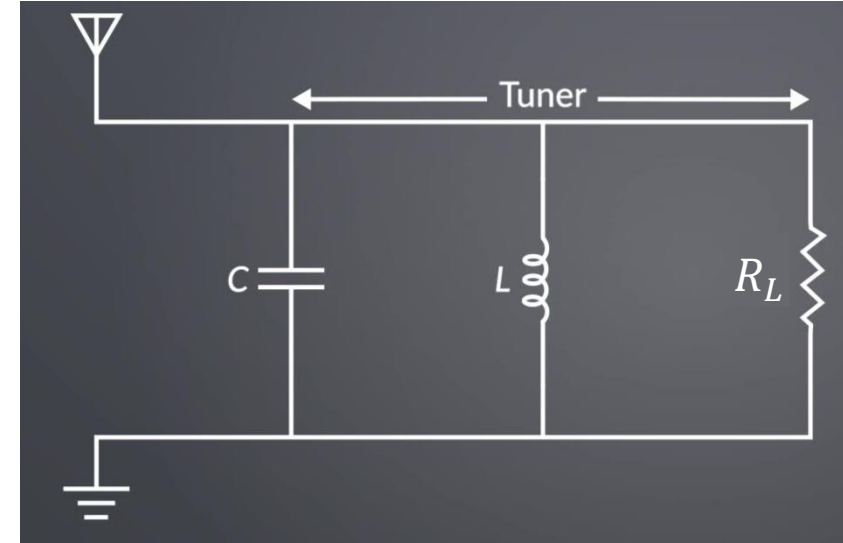
Therefore, impedance at resonance

$$Z \cong \frac{X_L^2}{R_L}$$

$$\Rightarrow Z \cong \frac{(2\pi f L)^2}{R_L}$$

$$\Rightarrow Z \cong \left[\frac{2\pi \left(\frac{1}{2\pi \sqrt{LC}} \right) L}{R_L} \right]^2$$

$$\Rightarrow Z \cong \frac{L}{CR_L}$$





Self Study :

Question: A coil of inductance L (including its internal resistance R) is connected in parallel with a capacitor of capacitance C . Derive an expression for the bandwidth of the parallel resonant circuit in terms of g and C .

Answer :

$$\Delta\omega = \omega_2 - \omega_1 = \frac{g}{C}$$

The Q of a parallel Circuit :

$$Q_p = \frac{\omega_m}{\Delta\omega} = \frac{\omega_m C}{g} = \frac{1}{g\omega_m L} = \frac{1}{g} \sqrt{\frac{C}{L}}$$

EXAMPLE 20.4

- Determine the Q_s and bandwidth for the response curve in Fig. 20.20.
- For $C = 101.5$ nF, determine L and R for the series resonant circuit.
- Determine the applied voltage.

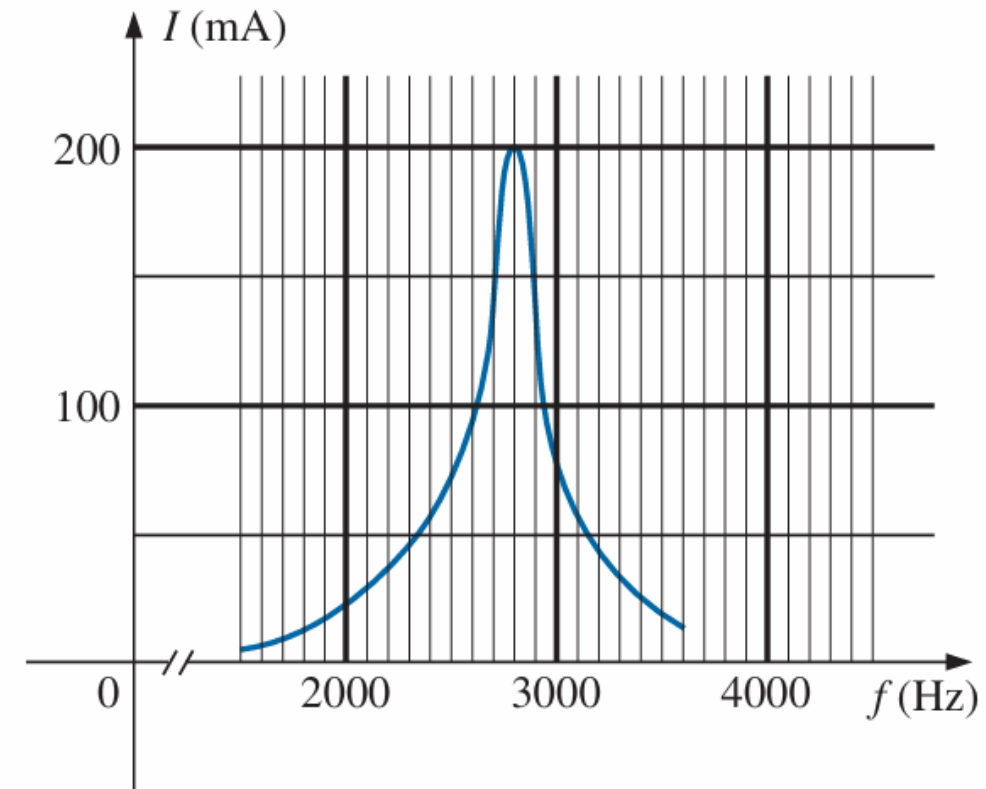


FIG. 20.20

Example 20.4

Parallel Resonance



EXAMPLE 20.6 Given the parallel network in Fig. 20.32 composed of “ideal” elements:

- Determine the resonant frequency f_p .
- Find the total impedance at resonance.
- Calculate the quality factor, bandwidth, and cutoff frequencies f_1 and f_2 of the system.
- Find the voltage V_C at resonance.
- Determine the currents I_L and I_C at resonance.

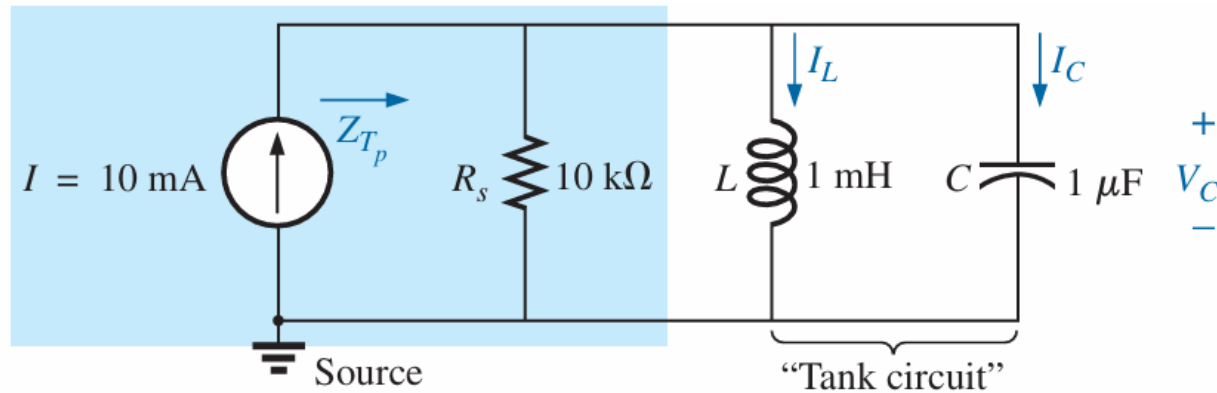


FIG. 20.32
Example 20.6.

EXAMPLE 20.11 Design a parallel resonant circuit to have the response curve in Fig. 20.38 using a 1 mH, $10\ \Omega$ inductor and a current source with an internal resistance of $40\ \text{k}\Omega$.

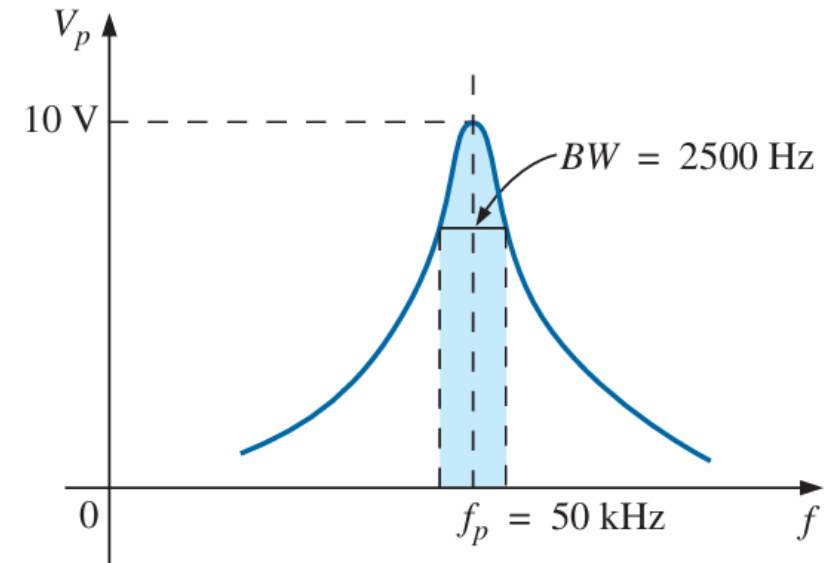
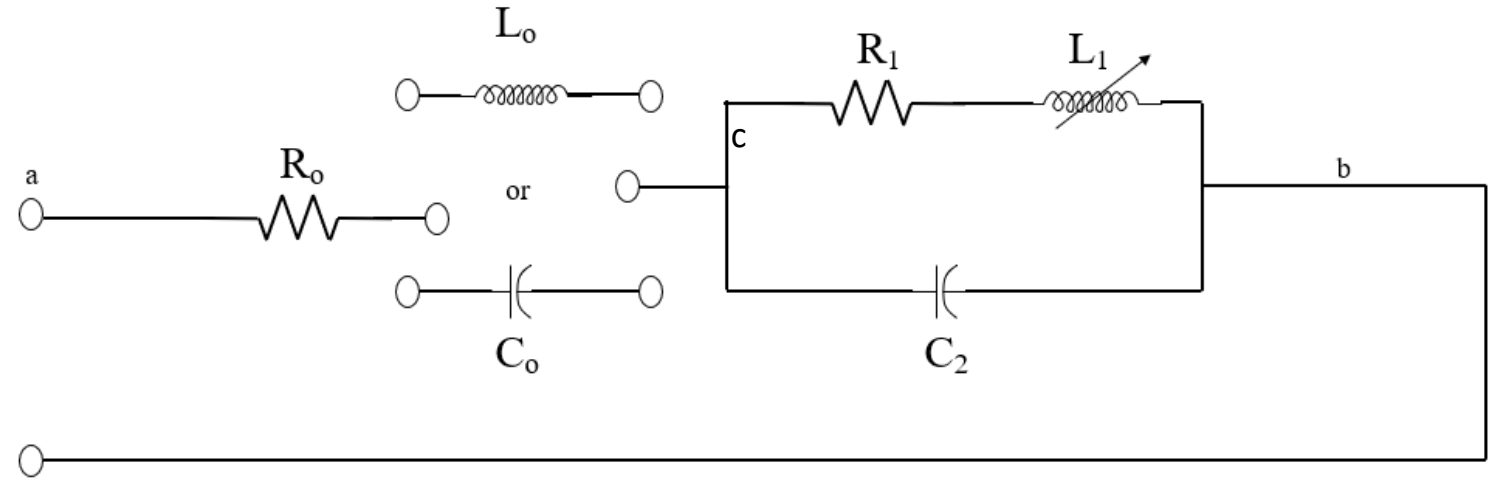


FIG. 20.38
Example 20.11.

Series Parallel Resonance



Question: The circuit of the following Fig. is to pass a 45 kHz current and is to block a 15 kHz current as effectively as possible. Calculate the value of L_1 and L_0 or C_0 . Given $R_0 = 20\Omega$, $R_1 = 40\Omega$ and $C_2 = 0.05\mu\text{F}$. Also find the impedance of L_1 for both 45 kHz and 15 kHz. Further, evaluate the impedance Z_{ab} for both 45 kHz and 15 kHz conditions.



Solution:

For the parallel resonance at 15KHz,

$$\begin{aligned}
 b_L &= b_C \\
 \Rightarrow \frac{\omega_p L}{\omega_p L_1 + R_1} &= \omega_p C_2 \\
 \Rightarrow \frac{L_1}{C} &= \omega_p^2 L_1 + R_1^2 \\
 \Rightarrow (2\pi \times 15000)^2 L_1 - \left(\frac{1}{0.05 \times 10^{-6}}\right) L_1 + 40^2 &= 0 \\
 \Rightarrow L_1 &= 2.1685 \text{ mH} \quad \text{or} \quad 83.06 \mu\text{H}
 \end{aligned}$$

We should select 2.1685mH for lower conductance

For the series resonance at 45KHz,

$$\begin{aligned}
 X_L &= 2\pi f_s L_1 = 2\pi \times 45000 \times 2.1685 \times 10^{-3} = 613.12 \Omega \\
 X_C &= \frac{1}{2\pi f_s C_2} = \frac{1}{2\pi \times 45000 \times 0.05 \times 10^{-6}} = 70.73 \Omega \\
 Z_{bc} &= (40 + j613.12) \parallel (-j70.73) = (0.676 - j79.9) \Omega
 \end{aligned}$$

For the series resonance at 45KHz, 79.9Ω inductive reactance must be placed in series with R_0 .

Series Parallel Resonance



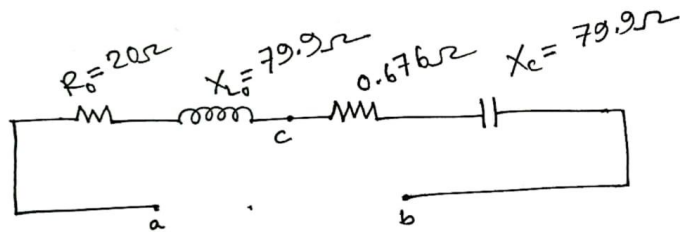
$$X_{L_0} = 2\pi f_s L_0$$

$$\Rightarrow L_0 = \frac{X_{L_0}}{2\pi f_s}$$

$$\Rightarrow L_0 = \frac{79.9}{2\pi \times 4500}$$

$$\Rightarrow L_0 = 0.282 \text{ mH}$$

The impedance at 45KHz,

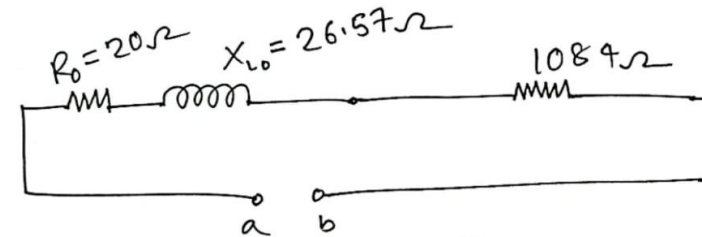


The actual impedance of $Z_{ab} = 20 + 0.676 = 20.676\Omega$

The impedance at 15KHz,

$$Z_{bc} = R_p = \frac{\omega^2 L_1^2 + R^2}{R_L} = \frac{(2\pi \times 15000 \times 2.1685 \times 10^{-3})^2 + 40^2}{40} = 1084\Omega$$

$$X_{L_0} = 2\pi f_s L_0 = 2\pi \times 15000 \times 0.282 \times 10^{-3} = 26.57\Omega$$



The actual impedance of $Z_{ab} = 20 + j26.57 + 1084$
 $= 1104 \angle 1.37^\circ \Omega$

Series Parallel Resonance



Assignment:

The circuit of the following Fig.33 is to pass a 10 kHz current and is to block a 20 kHz current as effectively as possible. Assume L_1 to have 0.005 H inductance and 50Ω resistance. Compare the circuit current at 10 kHz and 20 kHz operating conditions.

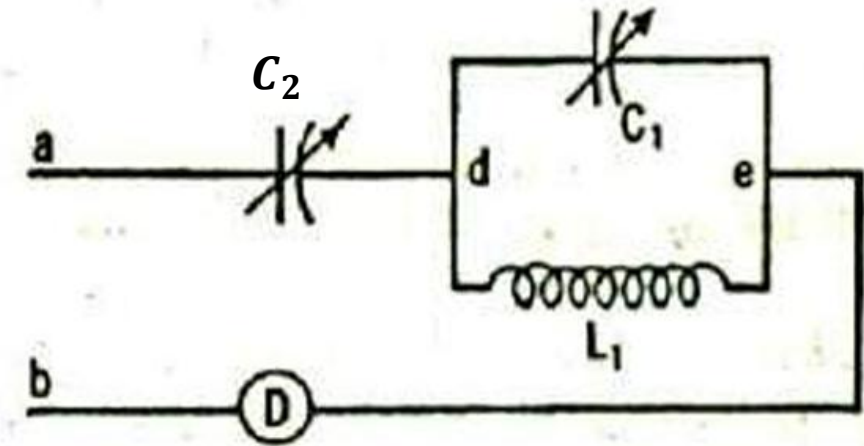


FIG. 33. Series-parallel tuning circuit.

Homework:

From: *Alternating Current Circuits* (4th edition)
Chapter V(Sinusoidal single-phase circuit Analysis)

- Example:7,9
- problems: 43,44,49,57

From: *Introductory Circuit Analysis* (11th edition)
Chapter 20(Resonance), **Only examples**

From: *Fundamentals of Electric Circuits* (5th edition)
Example:14.8,14.9 and Practice Problem: 14.8,14.9